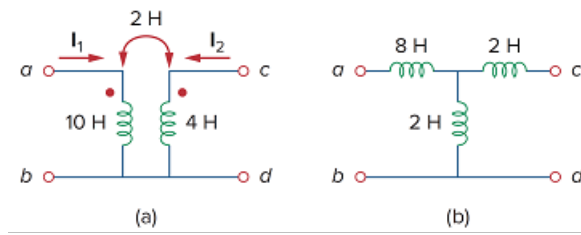


Problem

For the linear transformer in Fig. 13.26(a), find the Π equivalent network.

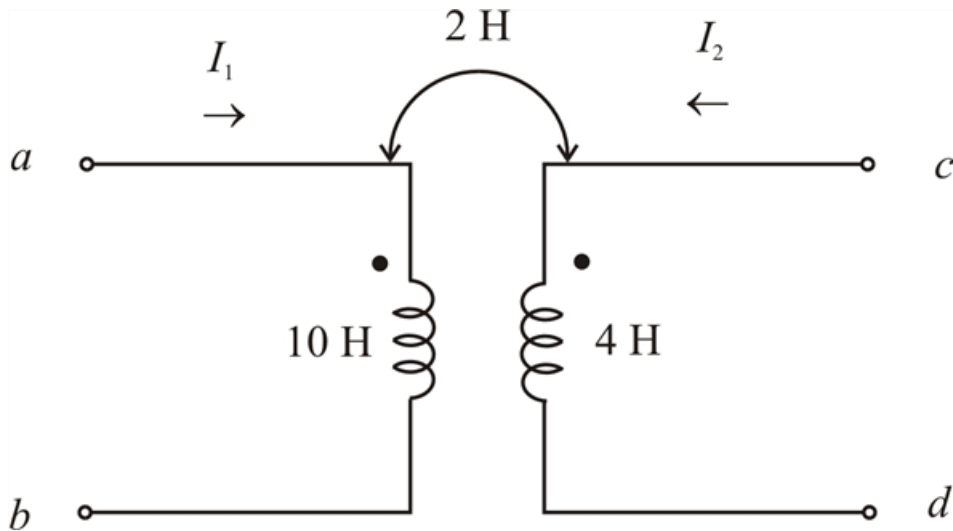
Figure 13.26: (a) a linear transformer, (b) its T-equivalent circuit.



Step-by-step solution

Step 1 of 4

Given circuit is,



Step 2 of 4

Given linear transform parameters are

$$L_1 = 10\text{H}, \quad L_2 = 4\text{H} \quad \text{and} \quad M = 2\text{H}$$

Thus, the Π network has the parameters are,

$$L_A = \frac{L_1 L_2 - M^2}{L_2 - M}$$

$$L_A = \frac{(10)(4) - (2)^2}{4 - 2}$$

$$L_A = \frac{40 - 4}{2}$$

$$\therefore \boxed{L_A = 18\text{ H}}$$

Step 3 of 4

$$L_B = \frac{L_1 L_2 - M^2}{L_1 - M}$$

$$L_B = \frac{(10)(4) - (2)^2}{10 - 2}$$

$$L_B = \frac{40 - 4}{8}$$

$$\therefore \boxed{L_B = 4.5 \text{ H}}$$

Step 4 of 4

$$L_C = \frac{L_1 L_2 - M^2}{M}$$

$$L_C = \frac{(10)(4) - (2)^2}{2}$$

$$L_C = \frac{40 - 4}{2}$$

$$\therefore \boxed{L_C = 18 \text{ H}}$$